

Appendix: Noken variations

Fix $i \in [m]$, for probabilities $\{p_{ij}\}_{j=1}^m$

define average Renyi α -entropy

$$E = E(\{p_{ij}\}) = \frac{1}{m} \sum_{i=1}^m H_{\alpha}^i$$

$$\text{where } H_{\alpha}^i = H_{\alpha}(\{p_{ij}\}_{j=1}^m).$$

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Input target token t_i from context $[t_1, \dots, t_m] \in \mathbb{R}^{r \times m}$

$$\text{Let } \alpha_{ij} = \text{softmax}_j \text{ score}(q_i, s_{k_j})$$

$$\text{and } j_i^{\alpha} = \arg \max_j \alpha_{ij}$$

$$\text{Define } h_i = \sum_{j=1}^m \alpha_{ij} v_j \in \mathbb{R}^n$$

$$\text{Let } \beta_{ij} = \text{softmax}_j \text{ score}(h_i, t_{k_j})$$

$$\text{and } j_i^{\beta} = \arg \max_j \beta_{ij}$$

$$\text{Define } q_i' = q_{j_i^{\beta}} \quad (0)$$

$$\text{Let } E_{q_i} = E(\{\beta_{ij}\}), \text{ similarly } E_{q_i'}.$$

If $E_{q'_i} < E_{q_i}$ replace q_i with q'_i and continue updating q_i while average Renyi α -entropy declines or is small enough.

Output token $o_{i+1} = t_{j_i^\beta} \in \mathbb{R}^r$.

Variation 1:

Define $q'_i = \sum_{j=1}^m \beta_{ij} q_j$ (instead of $q_{j_i^\beta}$). (1)

Variation 2:

Key $s_{k_{j_i^\alpha}}$ is most similar with respect to score

to a key $t_{k_{i^*}} \in \{t_{k_1}, \dots, t_{k_m}\}$.

Define $q'_i = q_{i^*}$. (2)

Let $E_{q_i} = E(\{\alpha_{ij}\})$, similarly $E_{q'_i}$

As before, if $E_{q'_i} < E_{q_i}$ update q_i and continue.

Output token $o_{i+1} = t_{j_i^\alpha} \in \mathbb{R}^r$.

Variation 3:

Let $\gamma_{ij} = \text{softmax score}(v_{j_i^\alpha}, t_{k_j})$

and $j_i^\gamma = \arg \max_j \gamma_{ij}$.

Define $q'_i = \sum_{j=1}^m \gamma_{ij} q_j$ (or $q_{j_i^\gamma}$). (3)

Let $E_{q_i} = E(\{\gamma_{ij}\})$, similarly $E_{q'_i}$.

As before if $E_{q'_i} < E_{q_i}$ update q_i with q'_i

and continue until average Renyi α -entropy

is acceptable.

Output token $o_{i+1} = t_{j_i^\gamma} \in \mathbb{R}^r$.

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Variation 4:

Let $\delta_{ij} = \text{softmax score}(q_i, v_j)$

and $j_i^\delta = \arg \max_j \delta_{ij}$.

Let $\gamma'_{ij} = \text{softmax score}(v_{j_i^\delta}, t_{K_j})$

and $j_i^{\gamma'} = \arg \max_j \gamma'_{ij}$.

(4) Define $q'_i = \sum_{j=1}^m \gamma'_{ij} q_j$ (or $q_{j_i^{\gamma'}}$). $o_{i+1} = t_{j_i^{\gamma'}}$

As before update while entropy is reduced.

Variation 5: cf variation 1

Let $h'_i = \sum_{j=1}^m \delta_{ij} v_j \in \mathbb{R}^n$,

and $\beta'_{ij} = \text{softmax score}(h'_i, t_{K_j})$

and $j_i^{\beta'} = \arg \max_j \beta'_{ij}$.

(5) Define $q'_i = \sum_{j=1}^m \beta'_{ij} q_j$ (or $q_{j_i^{\beta'}}$). $o_{i+1} = t_{j_i^{\beta'}}$

As before update while entropy is reduced.

Variation 6:

Key s_{j_i} is most similar with respect to score

a key $t_{k_{i^*}} \in \{t_{k_1}, \dots, t_{k_m}\}$

Define $q'_i = q_{i^*}$. (6)

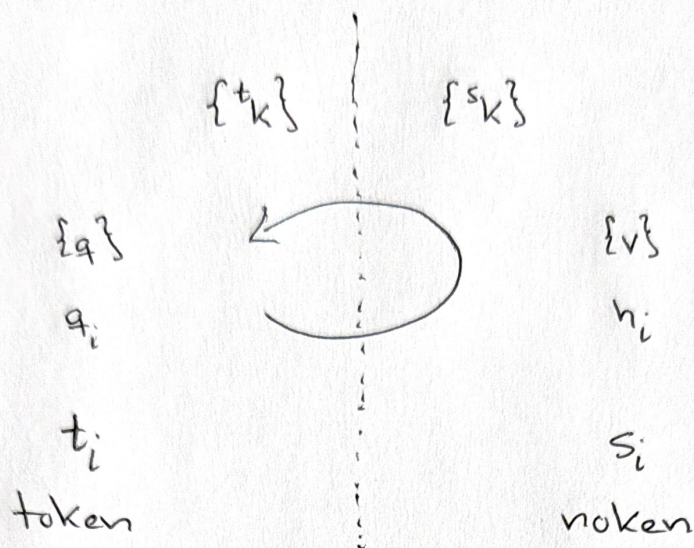
Let $E_{q_i} = E(\{s_{i,j}\})$ and $E_{q'_i}$ similarly.

If $E_{q'_i} < E_{q_i}$ assign $q_i := q'_i$

and output $o_{i+1} = t_{j_i}$.

This is a riff on Variation 4 and Variation 2.

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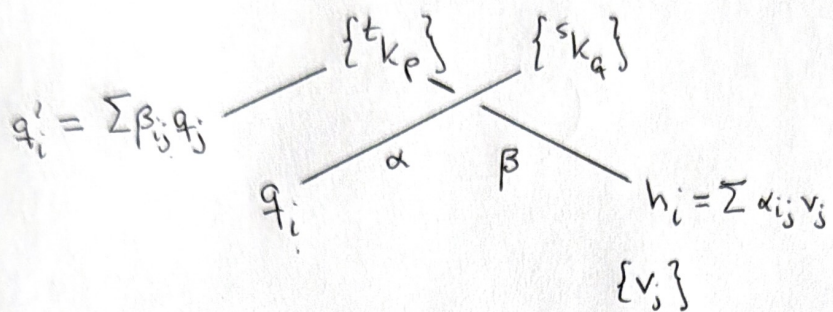


Score enables many comparisons to update q_i .

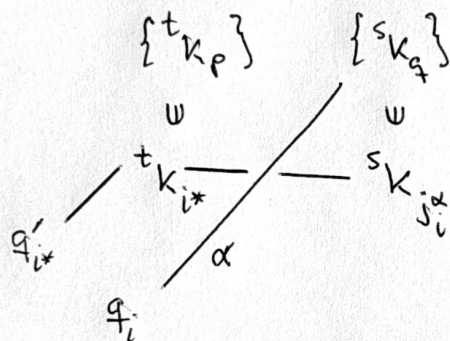
We have explored a few. Scaling and empirical evidence will reveal better choices.

We now summarize these noken variations by a catalogue of diagrams.

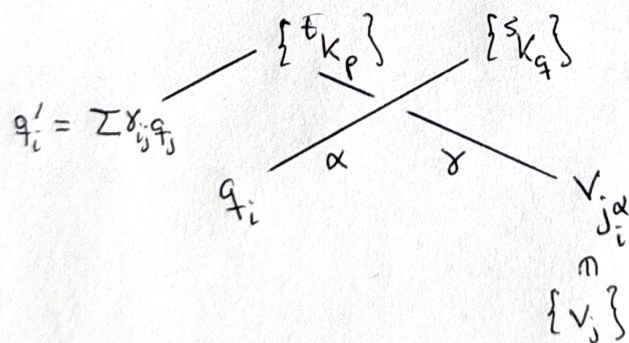
Variation 1:



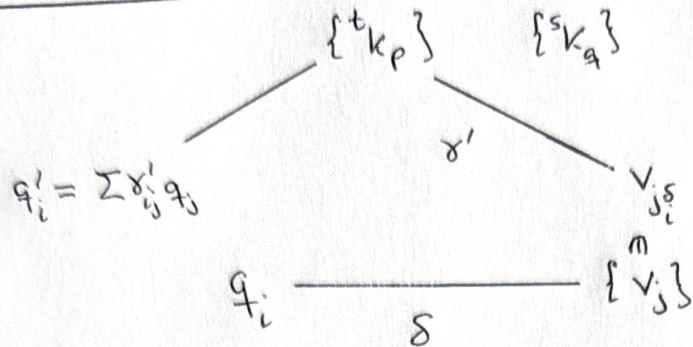
Variation 2:



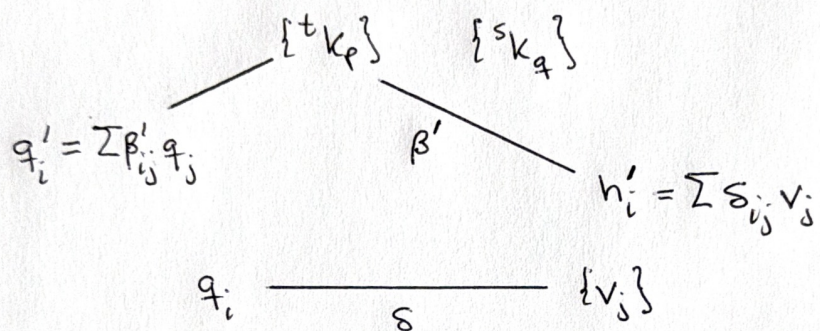
Variation 3:



Variation 4:



Variation 5:



Variation 6:

